

THE UNIVERSITY OF MELBOURNE

FNCE 30007 Derivative Securities

Lecture 9 – Dividends

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Lecture Outline

Options and dividend yields

- Options on stock indices
- Options on foreign exchange

Lower bounds for option prices

Put-call parity

Binomial model with dividend payment

- Binomial with dividend yield
- Binomial with known dollar dividend

BSM with dividend payment

- BSM with dividend yield
- BSM with known dollar dividend

Portfolio Insurance

Readings: Chapters 12, 13, 18

Options and Dividend Yields

How do we treat a stock which pays a known dividend yield?

- Take two identical stocks. Stock 1 pays no dividends, and stock 2 pays a dividend yield at rate q p.a.
- Both stocks will give an identical overall return. The return from stock 1 will be in the form of capital gains, and the return from stock 2 will be in the form of capital gains and dividends.
- The payment of a dividend will cause stock 2's price to drop by an amount equal to the value of the dividend.
- So payment of a dividend yield at rate q causes the growth rate to be less than that it would otherwise be, by an amount of q.

Options and Dividend Yields

Thus if the price of a stock, with a dividend yield of q, grows from S_0 today to S_T at time T:

- In the absence of dividends it would grow from S_0 today to $S_T e^{qT}$ at time T; or,
- In the absence of dividends it would grow from $S_0 e^{-qT}$ today to S_T at time T.

Therefore, we get the same probability distribution for the stock price at time T in each of the following cases:

- The stock price starts at price S_0 and pays a dividend yield at rate q; and,
- The stock price starts at price $S_0 e^{-qT}$ and pays no dividend yield.

Options and Dividend Yields

The rule which stems from the above example is:

- We can value European options lasting for time T on a stock paying a known dividend yield at rate q by reducing the stock price to $S_0 e^{-qT}$, and then valuing the option as though the stock pays no dividend.

The above rule can be applied to any option on an underlying asset which pays a known dividend yield.

In this lecture we focus on two particular options:

- Options on Stock Indices; and,
- Options on Currencies.

Options and Dividend Yields

Stock indices paying a known dividend yield:

- Exchange traded stock indices are available in most exchanges around the world;
- In Australia, options on the ASX200 are traded on the Australian Securities Exchange; and,
- In this lecture, we assume that dividends on stock indices are paid in terms of a dividend yield. They can also be paid in terms of a dollar value.

Options and Dividend Yields

Options on currencies:

- A foreign currency is analogous to a stock paying a known dividend yield.
- The owner of the foreign currency option receives a yield equal to the risk-free rate of interest in the foreign currency.
- The options are traded on both OTC markets and on an exchange, although they are mainly traded in OTC markets.
- FX options can be either European or American.

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Lower Bounds for Option Prices

The payment of a dividend by the underlying asset, lowers the price of that asset.

The reduction in price is good for the holder of a put option as it increases their payoff, while it is bad news for the holder of a call option as it reduces their payoff.

Thus, the lower bounds for both call and put options are affected by the payment of dividends.

Lower Bounds for Option Prices

The effect on lower bounds for European calls:

- In lecture 4 we saw that the lower bound for a European call on a non-dividend paying stock was:

$$c \geq S_0 - Ke^{-rT}$$

- Substituting S_0e^{-qT} for S_0, the lower bound for a European call option on a stock which pays a dividend yield is:

$$c \geq S_0e^{-qT} - Ke^{-rT}$$

Lower Bounds for Option Prices

We can prove the above result directly by considering the following two portfolios:

- Portfolio A: one European call option plus an amount of cash equal to Ke^{-rT}
- Portfolio B: e^{-qT} shares with dividends being reinvested in additional shares.

Lower Bounds for Option Prices

- In portfolio A, if the cash is invested at the risk-free interest rate, it will grow to K at time T. If $S_T > K$, the call option is exercised at time T, and portfolio A is worth S_T.
- If $S_T < K$, the call option expires worthless, and the portfolio is worth K.
- Hence, at time T portfolio A is worth:

$$Max(S_T, K)$$

- Due to the reinvestment of dividends, portfolio B becomes one share at time T. Thus, it is worth S_T at this time.
- Therefore, portfolio A must be worth at least as much as portfolio B at time T.
- Hence:

$$c + Ke^{-rT} \geq S_0 e^{-qT}$$

$$or$$

$$c \geq S_0 e^{-qT} - Ke^{-rT}$$

Lower Bounds for Option Prices

The effect on lower bounds for European puts:

- In lecture 4 we saw that the lower bound for a European put on a non-dividend paying stock was:

$$p \geq Ke^{-rT} - S_0$$

- Substituting S_0e^{-qt} for S_0, the lower bound for a European put option on a stock which pays a dividend yield is:

$$p \geq Ke^{-rT} - S_0e^{-qT}$$

Lower Bounds for Option Prices

As with the call option, this can be proved directly by forming the following two portfolios:

- Portfolio C: one European put option plus e^{-qt} shares with dividends on the shares being reinvested in additional shares
- Portfolio D: an amount of cash equal to Ke^{-rT}

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Portfolio Insurance

Put-Call Parity

In order to incorporate the effect of dividend yields on put-call parity:

- In lecture 4 we saw that put-call parity for a European call and put option on a non-dividend paying stock was:

$$c + Ke^{-rT} = p + S_0$$

- Replacing S_0 with S_0e^{-qT} yields the following put-call parity relationship on a stock paying a dividend yield at rate q:

$$c + Ke^{-rT} = p + S_0e^{-qT}$$

Put-Call Parity

We can prove the above result directly by considering the following two portfolios:

- Portfolio A: one European call option plus an amount of cash equal to Ke^{-rT}
- Portfolio C: one European put option plus e^{-qT} shares with dividends on the shares being reinvested in additional shares.
- Both portfolios must be worth $max(S_T, X)$ at time T. They must therefore be worth the same today, and the put-call parity result above holds.

Put-Call Parity

The above put-call parity relationship will hold for all options on underlying assets which pay a dividend yield.

A slight alteration to the formula is necessary when considering foreign currency options. q is replaced by r_f, where r_f is the foreign interest rate, and r is the domestic interest rate.

Thus, the relationship becomes:

$$c + Ke^{-rT} = p + S_0 e^{-r_f T}$$

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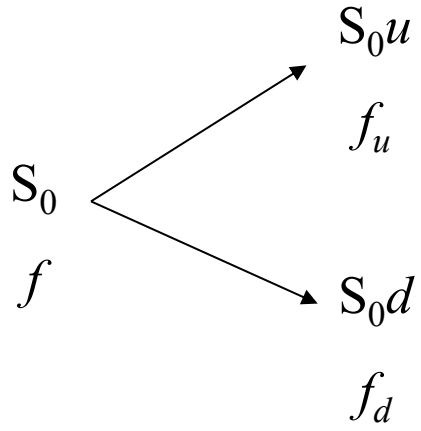
BSM with dividend payment

- BSM with dividend yield
- BSM with known dollar dividend

Portfolio Insurance

Binomial Model

To examine the effect of a dividend yield equal to q on the binomial model discussed in Lecture 5, consider the following tree diagram:



The stock price in this diagram starts at S_0 and either moves up to S_0u or down to S_0d.

Binomial Model

We define p as the probability of an up movement in a risk-neutral world.

The total return provided by the stock in a risk-neutral world must be the risk-free interest rate r.

Our stock pays dividends equal to a return of q.

Therefore, the return from capital gains must be equal to r-q.

p must satisfy:

$$pS_0u + (1-p)S_0d = S_0e^{(r-q)T}$$

$$or$$

$$p = \frac{e^{(r-q)T} - d}{u - d}$$

Binomial Model

As we saw in Lecture 5, the value of the option at time zero is the expected payoff in a risk-neutral world discounted at the risk-free rate:

$$f = e^{-rT}[pf_u + (1-p)f_d]$$

Binomial Model

Example:

- Consider a six-month put option on a stock with a strike price of $40. Suppose that the initial stock price is $50 and the stock price will either move up to $60 or down to $35 during a six-month period.
  - The risk-free interest rate is 6% p.a. and the stock is expected to provide a dividend yield of 2% p.a.

Binomial Model

- Solution:

$$u = 1.2 \text{ and } d = 0.7$$

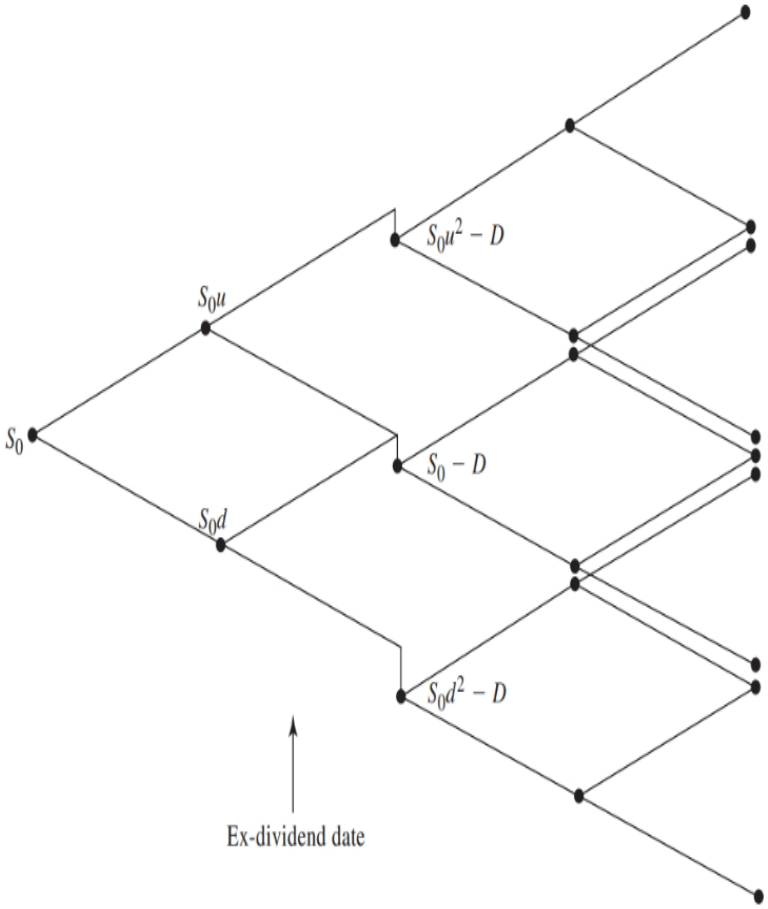
$$p = \frac{e^{(0.06-0.02)\times 6/12} - 0.7}{1.2 - 0.7} = 0.6404$$

If the stock price moves up the payoff is zero, if it moves down the payoff is \$5. The value of the option is therefore:

$$e^{-0.06\times 6/12}[0.6404 \times 0 + 0.3596 \times 5] = 1.74$$

Binomial Model

- Now assume the dividend of a known dollar amount will be paid on the underlying stock during the life of the option.
- Stock price will drop by the amount of dividend on the ex-dividend date.
- Tree will not recombine…
- So need to adjust the S_0 to make the tree recombine and then use the binomial mode.



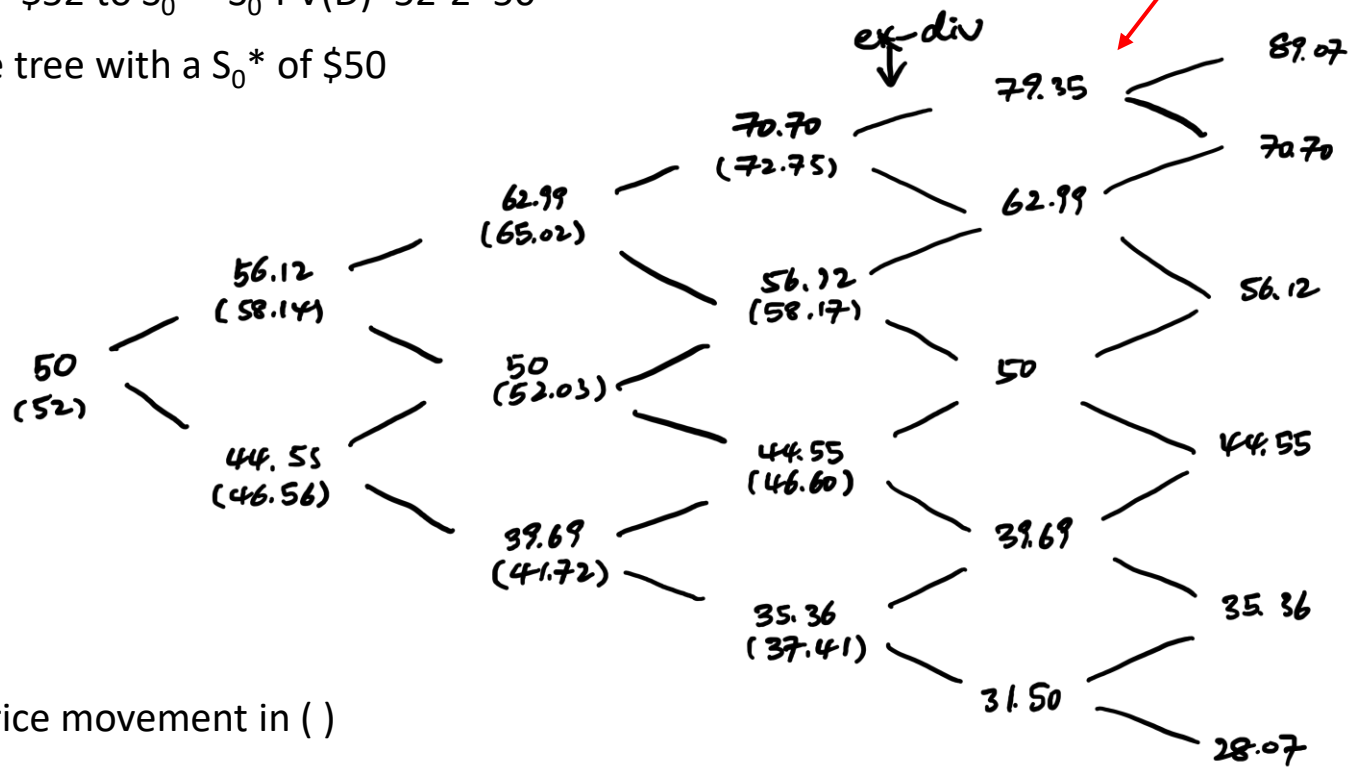
Binomial Model

An European put option with S_0=\$52, K=\$50, r=10%, u=1.1224, d=0.8909, T=5 months with 5 time steps.

Dividend of \$2.06 to be paid in 3.5 months.

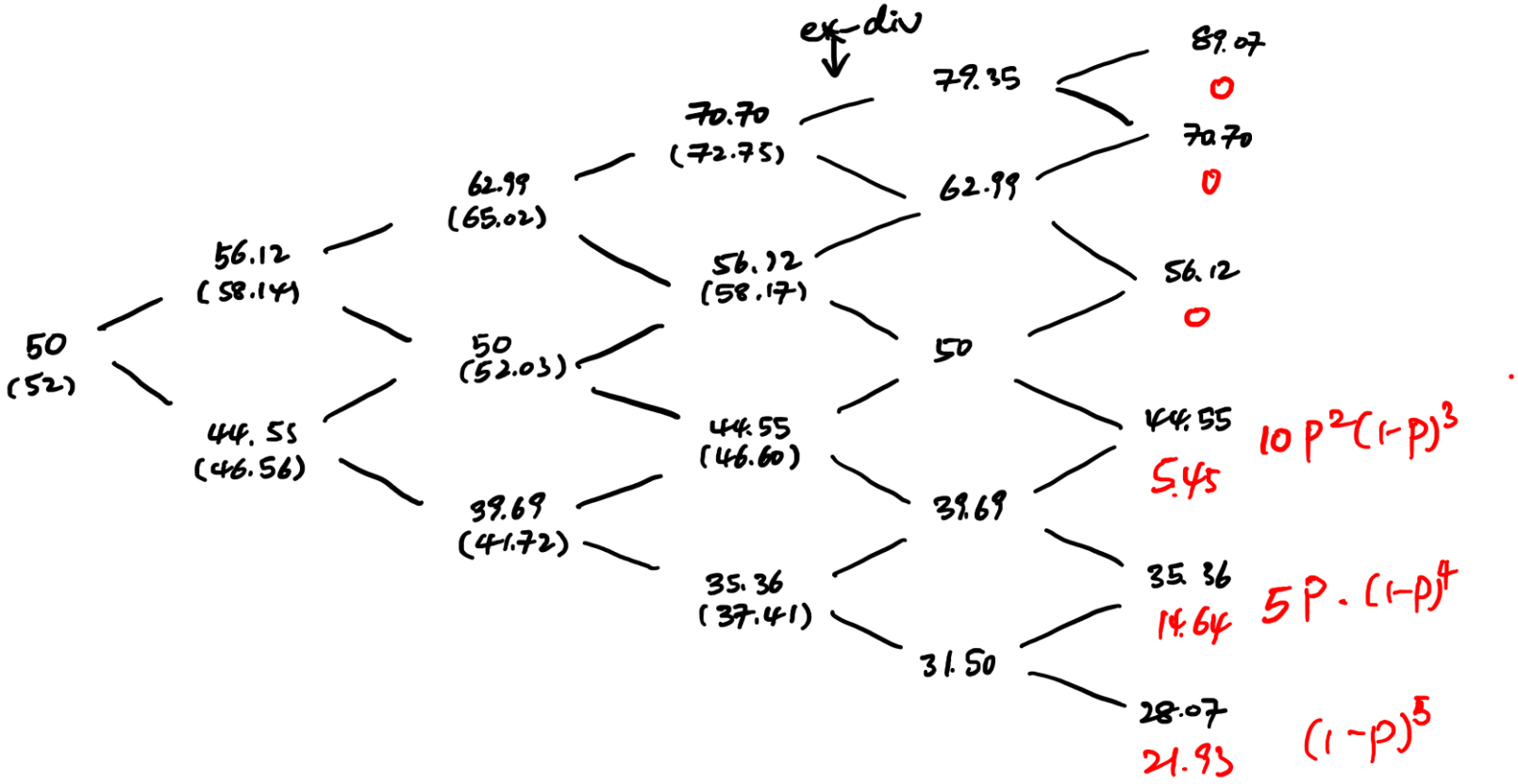
- PV of D=2.06*e^(-0.1*3.5/12)=2
- Adjust S_0=\$52 to S_0*= S_0-PV(D)=52-2=50
- Draw the tree with a S_0* of \$50

79.35=70.70*1.1224



- Actual price movement in ( )

Binomial Model



A 5-step binomial model is not examinable. You are not required to understand how the probability for each end note is calculated. This question is for illustration purpose only.

$$f = \frac{10p^2(1-p)^3 * 5.45 + 5p*(1-p)^4 * 14.64 + (1-p)^5 * 21.93}{e^{0.1*\frac{5}{12}}} \text{ where } p = \frac{e^{0.1*1/12} - 0.8909}{1.1224 - 0.8909} = 0.5073$$

$$f = 4.32$$

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- BSM with dividend in fixed dollar amount

Portfolio Insurance

BSM for Dividend Yielding Stocks

Substituting $S_0 e^{-qT}$ for S_0 in the BSM formulae we covered in lecture 8 yields the following formulae to price both European calls and puts on dividend yielding stocks:

$$c = S_0 e^{-qT} N(d_1) - K e^{-rT} N(d_2)$$

$$p = K e^{-rT} N(-d_2) - S_0 e^{-qT} N(-d_1)$$

$$\text{As:} \quad \ln \frac{S_0 e^{-qT}}{K} = \ln \frac{S_0}{K} - qT$$

$$d_1 = \frac{\ln(S_0/K) + (r - q + \sigma^2/2)T}{\sigma\sqrt{T}}$$

$$d_2 = \frac{\ln(S_0/K) + (r - q - \sigma^2/2)T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}$$

BSM for Dividend Yielding Stocks

A slight alteration is necessary to the previous BSM Model when considering FX options. q is replaced by r_f, where r_f is the foreign interest rate, and r is the domestic interest rate.

Black-Scholes becomes:

$$c = S_0 e^{-r_f T} N(d_1) - K e^{-rT} N(d_2)$$

$$p = K e^{-rT} N(-d_2) - S_0 e^{-r_f T} N(-d_1)$$

$$\text{As:} \quad \ln \frac{S_0 e^{-r_f T}}{K} = \ln \frac{S_0}{K} - r_f T$$

$$d_1 = \frac{\ln(S_0/K) + (r - r_f + \sigma^2/2)T}{\sigma\sqrt{T}}$$

$$d_2 = \frac{\ln(S_0/K) + (r - r_f - \sigma^2/2)T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}$$

BSM for Dividend Yielding Stocks

What if the dividend is a fixed dollar amount?

We can examine European options by assuming that there are two components to a stock's price:

- A risky component; and,
- A riskless component that is used to pay the known dividend.
- Thus the riskless component is equal to the PV of all dividends discounted from the ex-dividend dates to the present time at the risk-free rate.

BSM for Dividend Yielding Stocks

Therefore, BSM can be used as long as we replace S_0 (from the original BSM on non-dividend paying stock) by the risky component of the dividend-paying stock.

This can be done by subtracting the PV of the dividends from the S_0 of dividend-paying stock.

Remember, that we only include the dividend in the calculation if its ex-dividend date occurs during the life of the option.

BSM for Dividend Yielding Stocks

Therefore, formulas for European options become:

$$c = [S_0 - PV(D)]N(d_1) - Ke^{-rT}N(d_2)$$

$$p = Ke^{-rT}N(-d_2) - [S_0 - PV(D)]N(-d_1)$$

Where:

$$d_1 = \frac{\ln[(S_0 - PV(D))/K] + (r + \sigma^2/2)T}{\sigma\sqrt{T}}$$

$$d_2 = \frac{\ln[(S_0 - PV(D))/K] + (r - \sigma^2/2)T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}$$

BSM for Dividend Yielding Stocks

Consider a European call option on a stock with ex-dividend dates in three months and seven months. The dividend on each ex-dividend date is expected to be $0.75. The current share price is $30, the exercise price is $29, the stock price volatility is 25% per annum, the risk-free rate of interest is 8% per annum, and the time to maturity is nine months. What is the price of the call option?

BSM for Dividend Yielding Stocks

The call option has a price of:

$$0.75e^{-0.08x3/12} + 0.75e^{-0.08x7/12} = 1.451$$

$$d_1 = \frac{\ln((30-1.451)/29)+(0.08+0.25^2/2)\frac{9}{12}}{0.25\sqrt{\frac{9}{12}}} = 0.3130 \approx 0.31$$

$$d_2 = 0.3130 - 0.25\sqrt{\frac{9}{12}} = 0.09$$

$$c = (S_0 - \text{PV(D)})\,N(d_1) - Ke^{-rT}N(d_2)$$

$$= (30 - 1.451)N(0.31) - 29e^{-0.08\times 9/12}N(0.09)$$

$$= 28.549 \times (0.6217) - 29 \times (0.94176) \times (0.5359)$$

$$= \$3.11$$

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Portfolio Insurance

Portfolio Insurance

Portfolio managers can use index options as insurance against the value of their portfolios falling.

Consider a manager of a well diversified portfolio which mimics the ASX 200 Index, and as such has similar returns and dividend yield.

- The percentage changes in the value of the portfolio can be expected to be approximately the same as the percentage changes in the value of the index.
- We know that an index option on the ASX 200 is on 25 times the index, and as such, a long put option on the ASX 200 would allow a manager to hedge against the risk of the value of their portfolio falling.

Portfolio Insurance

Suppose that the manager's portfolio is worth $100 million, and the value of the index is currently 5,000. The volatility of the index is 25% p.a., the risk-free interest rate is 9% p.a., the manager wants to hedge the portfolio for the next 3 months, the dividend yield on the index is 5% p.a., and put options can be bought with an exercise price of 5,000.

- How many options would the manager need to purchase?
- How much would this insurance cost?
- Show that the strategy works?

Portfolio Insurance

1. The manager would need to buy put options on the ASX 200. This would give them a payoff on the option should the value of the index fall.
2. The manager would need to purchase the following number of put options:

$$\frac{\text{Value to be hedged}}{\text{Value one put option will hedge}} = \frac{100{,}000{,}000}{5000 * 25} = 800$$

Portfolio Insurance

The total cost of the insurance to protect the $100 million portfolio is calculated as follows:

$$d_1 = \frac{\ln(5000/5000) + (0.09 - 0.05 + 0.25^2/2)0.25}{0.25\sqrt{0.25}} = 0.1425 \approx 0.14$$

$$d_2 = d_1 - 0.25\sqrt{0.25} = 0.02$$

$$p = 5000e^{-0.09x0.25}\, N(-d_2) - 5000e^{-0.05x0.25} N(-d_1) = 211.36$$

Total cost=$211.36x800x25 = \$4{,}227{,}200$

Portfolio Insurance

To illustrate that the insurance works, consider the situation where the value of the index drops to 4,000 in three months.

- The portfolio will be worth:
  - 4,000/5,000 x $100 million = $80 million
- The payoff from the options will be:
  - 800 x (5000-4000) x 25 = $20 million
- This brings the total value of the portfolio up to the insured value of $100 million (or $95.77 million when the cost of the options is accounted for).

Practice Questions

An index currently stands at 1,500. European call and put options with a strike price of 1,400 and time to maturity of six months have market prices of 154.00 and 34.25, respectively. The six-month risk-free rate is 5%. What is the implied dividend yield?

The implied dividend yield is the value of q that satisfies the put–call parity equation. It is the value of q that solves

$$154 + 1400e^{-0.05\times0.5} = 34.25 + 1500e^{-0.5q}$$

This is 1.99%.

Practice Questions

Show that a European call option on a currency has the same price as the corresponding European put option on the currency when the forward price equals the strike price.

This follows from put–call parity and the relationship between the forward price, F_0, and the spot price, S_0.

$$c + Ke^{-rT} = p + S_0 e^{-r_f T}$$

and

$$F_0 = S_0 e^{(r-r_f)T}$$

so that

$$c + Ke^{-rT} = p + F_0 e^{-rT}$$

If $K = F_0$ this reduces to $c = p$. The result that $c = p$ when $K = F_0$ is true for options on all underlying assets, not just options on currencies. An at-the-money option is frequently defined as one where $K = F_0$ (or $c = p$) rather than one where $K = S_0$.

Next Week

Greek Letters, with a focus on DELTA